ALICE L WALTON SCHOOL OF MEDICINE

Spectrum of Activity and Indications 1 – β-lactams and Vancomycin

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Learning Objectives (2)

After completing this brick, you will be able to:

1. Describe the spectrum of activity of β-lactams, Vancomycin, and Bacitracin
2. Recognize the infection and choose the appropriate treatment agent.

CASE CONNECTION

AB, a 5-year-old child, presents with complaints of ear pain and irritability. The symptoms began three days ago, with AB frequently tugging at their right ear and experiencing difficulty sleeping. The parents report that AB has been more fussy than usual and has had a low-grade fever. There is no history of trauma to the ear, and AB has not had any recent upper respiratory infections. On examination, AB appears uncomfortable and is holding their right ear. The vital signs show a mild fever. Otoscopic examination reveals a red, bulging tympanic membrane in the right ear, with decreased mobility on pneumatic otoscopy. The left ear appears normal. There is no discharge from the ear, and no lymphadenopathy is noted. Based on the clinical presentation and examination findings, the diagnosis of Otitis Media is made.

What can you give AB to treat their condition? Consider your answer as you read, and we'll revisit AB at the end of the brick.

GO TO CONCLUSION ↓

In the cephalosporin mechanism of action brick, we have discussed the mechanism of action and adverse effects of two major drug classes b-lactams and vancomycin. B-lactams include the penicillins (including Amoxicillin and piperacillin-tazobactam), cephalosporins (all five generations), monobactams, and carbapenems.

ACTIVE LEARNING QUE[illegible]

Review: Match the antibiotic to the most appropriate drug class.

Nafcillin	Antistaphylococcal Penicillin
Cefepime	Cephalosporin (4th Generation)
Piperacillin-Tazobactam	Anti-pseudomonal Penicillin
Ceftriaxone	Cephalosporin (3rd Generation)

Explanation

Nafcillin is an anti-staphylococcal Penicillin. Some agents have developed resistance to this drug class. The most important example is Methicillin-resistant *Staphylococcus Aureus*. Ceftriaxone and Cefepime are third —and fourth-generation Cephalosporins. Piperacillin-tazobactam is a combination of b-lactamase inhibitors that has very broad coverage, including Pseudomonas aeruginosa; hence, it is called Anti-pseudomonal Penicillin.

This is our first 'coverage brick'. In this brick, we will discuss what organisms or bugs these agents car[illegible] *spectrum of activity or coverage of the antibiotic*. The spectrum of activity helps to choose antibiotics for empirical or targeted treatment of infection. Since you have learned about the Gram-positive organisms, we are going to first look at the spectrum of activity of penicillin against Gram-positive organisms. As we learn more about the Gram-negative and atypical organisms, we will discuss their coverage in the subsequent weeks. Often, when considering the spectrum of activity of antibiotics, it is helpful to organize coverage against classes of bacteria and certain drug-resistant organisms as follows: Streptococci, Enterococci, Staph species, MRSA, Gram Negative Rods, ESBLs and Pseudomonas, Neisseria, Anaerobes, and/ or atypicals.

What is Penicillin's coverage and what infections are they indicated for? (LO # 1 and 2)

The **penicillins (Penicillin VK and G)** have Gram-positive cocci coverage. These would include Staphylococci and some Streptococci. However, many of these organisms have developed resistance to penicillin. However, oral penicillin (Penicillin VK) has been useful as a first line agent in the treatment of many streptococcal infections. Other antibiotic options (as 2nd line) are also available, we will discuss these later as we learn more agents.

Antistaphylococcal penicllins (Nafcillin, Oxacillin) were developed to overcome the resistance against Staphylococcal resistance. Strains of Staphylococcus aureus are susceptible to these agents. Needless to say, the Methicillin-resistant Staphylococcus aureus (MRSA) strains are not susceptible to these agents.

What is the primary use of antistaphylococcal penicillins like Nafcillin and Oxacillin?

Broad-spectrum penicillins such as **Amoxicillin and Ampicillin** have good gram-positive and gram-negative coverage (will discuss in week 2, session 2 brick). Amoxicillin has broad Gram-positive and -negative coverage (discussed in week 2, session 2).

Piperacillin-tazobactam is a combination antibiotic that provides broad-spectrum coverage against a wide range of bacteria, including Gram-positive, Gram-negative, and anaerobic organisms. Since it is an antipseudomonal penicillin, it is particularly effective against *Pseudomonas aeruginosa*. The addition of tazobactam, a β-lactamase inhibitor, enhances the efficacy of piperacillin by protecting it from degradation by β-lactamase enzymes produced by certain bacteria.

What is the role of tazobactam in piperacillin-tazobactam?

What are Cephalosporins' coverage and what infections are they indicated for? (LO # 1 and 2)

First-generation cephalosporins, such as cephalexin (PO) and cefazolin (IV), are primarily effective against Gram-positive bacteria, including *Staphylococcus aureus* and Streptococcus species. They have *limited activity against Gram-negative bacteria* and are generally well-tolerated. While both antibiotics have good covers Gram-positive coverage, they do NOT cover MRSA.

Second-generation cephalosporins, such as cefuroxime, cefoxitin, and cefotetan, have very limited Gram-positive coverage and are more effective against Gram-negative bacteria (covered in more detail in the upcoming weeks). Unlike extended spectrum penicillins, cephalosporins do not cover Bacteroides species well with the exception of two second generation cephalosporins: cefoxitin and cefotetan.

What infections are first-generation cephalosporins commonly used to treat?

Third-generation cephalosporins, such as ceftriaxone and ceftazidime, have a broad spectrum of activity against both Gram-positive bacteria, and their Gram-negative coverage is extended compared to first and second generations. This includes coverage of *Neisseria meningitidis* and *Neisseria gonorrhea*.

Fourth-generation cephalosporins, like cefepime, retain good Gram-positive coverage (except MRSA) but have increased Gram-negative coverage compared to third-generations (discussed in week 2).

Fifth-generation cephalosporins, such as ceftaroline, are notable for their activity against methicillin-resistant Staphylococcus aureus (MRSA) while maintaining broad-spectrum activity against Gram-negative bacteria.

Although many cephalosporins cover gram positive cocci, Enterococcus species are known for their intrinsic resistance to cephalosporins.

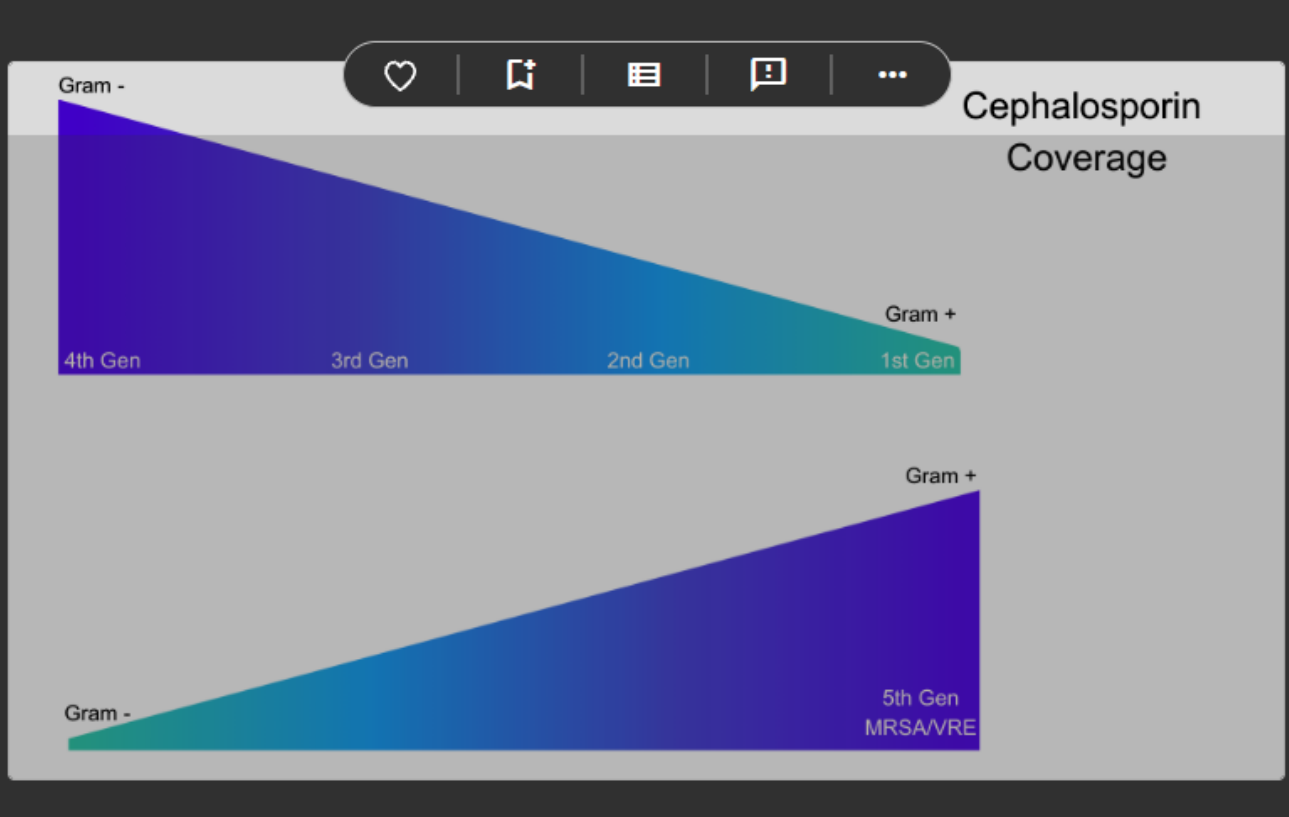

Figure 1

What is the primary advantage of fifth-generation cephalosporins like ceftaroline?

What is Vanocmycin's coverage and what infections are they indicated for? (LO # 1 and 2)

Vancomycin is effective against most Gram-positive organisms. It does not have good Gram-negative coverage. Vancomycin is one of the few antibiotics that is effective against MRSA. Vancomycin-resistant Enterococcus (VRE) refers to strains of the Enterococcus bacteria that have developed resistance to vancomycin. In the other bricks, we discussed *C. difficile*-associated diarrhea; vancomycin is one of the few antibiotics that is used in the treatment of *C. difficile* infection. Vancomycin (oral) is the drug-of-choice used in the treatment of *Clostridium difficile* infections.

CLINICAL CORRELATION

Can you use IV vancomycin to treat a C. difficile infection? No! Let's take a step back to some of our pharmacokinetics...

When vancomycin is administered orally it is not readily absorbed through the mucosa of the gastric mucosa, essentially the bioavailability is zero because it is not absorbed. On the other hand, IV vancomycin has great bioavailability to treat many systemic infections (duh... an IV drug's bioavailability is basically 100%!), BUT it does not cross into the gastric mucosa. Since, C. difficile infections reside primarily in the gastric mucosa, ORAL vancomycin acts as a topical treatment and IV vancomycin would be ineffective in the case.

What is Bacitra[illegible]fections are they indicated for? (LO # 1 and 2)

Bacitracin is effective against Gram-positive bacteria, including Staphylococcus aureus and Streptococcus species. It works by inhibiting cell wall synthesis, making it useful as a topical treatment for treating skin infections caused by minor cuts, scrapes, or burns.

What is the primary use of oral vancomycin?

CLINICAL CORRELATION

Bacitracin is a key component in triple antibiotic ointments, which are commonly used for preventing infections in minor cuts, scrapes, and burns. It works by inhibiting bacterial cell wall synthesis, making it effective against Gram-positive bacteria such as Staphylococcus aureus and Streptococcus species. When combined with othe[illegible], bacitracin provides broad-spectrum coverage, enhancing the ointment's ability to prevent infections from a wider range of bacteria. This combination is particularly useful in topical applications due to its effectiveness and low risk of systemic toxicity.

Table: Summary Table of Infections and First-line Antibiotics

Infection	First-line Antibiotic
Pharyngitis (typically due to Strep species)	Oral Penicillins
Otitis Media (typically due to Strep pneumonia and susceptible GN bacteria)	Amoxicillin (with or without clavulanic acid)
Uncomplicated Skin and Soft Tissue Infections (typically due to GPCs)	First-generation Cephalosporins (e.g., cephalexin)
Surgical Prophylaxis of procedures through the skin, typically covering GPCs with an IV agent	Cefazoline
MRSA Infections	Vancomycin, Ceftarolene
C. diff Infections	Vancomycin (oral)

CASE CONNECTION

BACK TO INTRODUCTION ↑

Based on your reading, what would you recommend for AB?

AB has symptoms lasting > 48 hours, so would be prescribed a course of antibiotics to treat the suspected bacterial infection. When antibiotics are indicated for Otitis Media, typical choices to cover Strep and susceptible gram negative bacteria include amoxicillin or amoxicillin-clavulanate. It's important for AB to complete the f[illegible] improve, before finishing the medica[illegible]nter pain relievers such as acetaminophen or ibuprofen (oral liquid dosage) can help manage the discomfort and reduce fever. The parents should follow the dosing instructions provided by the healthcare provider. Ensuring that AB gets plenty of rest and stays well-hydrated can help support the body's healing process.

Summary

- **Penicillins**: Effective against Gram-positive cocci, including Staphylococci and some Streptococci. However, many organisms have developed resistance. Oral penicillins are useful as first-line agents for treating pharyngitis which is typically due to Strep pneumonia.
- **Antistaphylococcal Penicillins:** Nafcillin and Oxacillin were developed to overcome Staphylococcal resistance. They are effective against Staphylococcus aureus but not Methicillin-resistant Staphylococcus aureus (MRSA).
- **Broad-Spectrum Penicillins:** Amoxicillin and Ampicillin have good Gram-positive and Gram-negative coverage.
- **Piperacillin-Tazobactam:** Provides broad-spectrum coverage, including Pseudomonas aeruginosa.
- **Cephalosporins:** Different generations have varying coverage. First-generation (e.g., cephalexin) is effective against Gram-positive bacteria. Third-generation (e.g., ceftriaxone) has broad-spectrum activity including Neisseria. Fifth-generation (e.g., ceftaroline) is effective against MRSA.
- **Vancomycin:** Effective against most Gram-positive organisms and MRSA. Used for treating C. difficile infections and infections caused by Vancomycin-resi[illegible]
- **Bacitracin:** Effective against Gram-positive bacteria, used for treating minor skin infections.

Review Questions

1. Which first-generation cephalosporin is commonly used for coverage of gram positive cocci for surgical prophylaxis? It needs to be delivered intravenously for rapid bioavailability and peak tissue levels at the time of the surgical incision.

A. ○ Cephalexin
B. ○ Cefuroxime
C. ○ Amoxicillin
D. ● Cefazolin

Hide Explanation

The correct answer is cefazolin (D). Cefazolin is commonly used for surgical prophylaxis due to its effectiveness against Gram-positive bacteria and some Gram-negative bacteria and its rapid bioavailability by IV administration. Cephalexin (A) is also a first-generation cephalosporin but oral medications are not typically used for surgical prophylaxis. Cefuroxime (B) is a second-generation cephalosporins, which have different indications and coverage. Amoxicillin (C) is not a first-generation cephalosporin.

2. Which penicillin is used as a first-line agent for Otitis Media?

A. ● Amoxicillin
B. ○ Nafcillin
C. ○ Oxacillin
D. ○ Penicillin G

Hide Explanation

The correct answer is Amoxicillin (A). Amoxicillin (A) and its combination with clavulanic acid are used as first-line agents in the treatment of Otitis Media. Nafcillin (B) and Oxacillin (C) are antistaphylococcal penicillins, not typically used for Otitis Media. Vancomycin (C) can be used for respiratory tract infections but is not the first-line treatment for Otitis Media. Penicillin G (D) is not the first-line treatment agent for Otitis Media.

3. Which cephalosporin is effective against MRSA?

A. ○ Cefepime
B. ● Ceftaroline
C. ○ Ceftriaxo[illegible]
D. ○ Cefazolin

Hide Explanation

The correct answer is Ceftaroline (B). Ceftaroline is a fifth-generation cephalosporin known for its activity against MRSA. Cefepime (A) is a fourth-generation cephalosporin and does not cover MRSA. Ceftriaxone (C) is a third-generation cephalosporin with no MRSA coverage. Cefazolin (D) is a first-generation cephalosporin and also does not cover MRSA.

4. A patient was diagnosed with bacterial pharyngitis from *Streptococcus pyogenes*. What would be the first-line treatment for this patient?

A. ○ Nafcillin
B. ○ Vancomycin
C. ○ Cefuroxime
D. ● Penicillin VK

Hide Explanation

The correct answer is Gram-positive cocci (A). Penicillin are effective against Staphylococci and some Streptococci. Nafcillin (B) covers *Staphylococcus*, while *Streptococcus* are re[illegible] cover gram-positive

Hide Explanation

The correct answer is Gram-positive cocci (A). Penicillin are effective against Staphylococci and some Streptococci. Nafcillin (B) covers *Staphylococcus*, while *Streptococcus* are resistant to Nafcillin. While Vancomycin does cover gram-positive bacteria, it is not the first-line treatment for pharyngitis. Cefuroxime, 2nd-generation cephalosporin, is not the first-line treatment for pharyngitis.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the spectrum of activity of β-lactams, Vancomycin, and Bacitracin~~
02 ☑ ~~Recognize the infection and choose the appropriate treatment agent.~~

Objective Confidence: 2/2